

Fig. 5: Master and slave connection

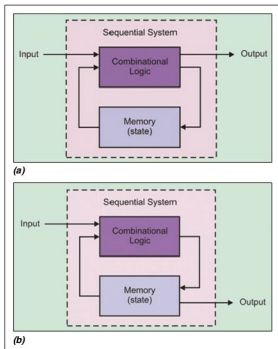


Fig. 6: (a) Mealy and (b) Moore FSM

for connecting the master and the slave devices in an I²C protocol. Multiple slave devices can be connected using the same two lines.

I²C master controller in an FPGA will be implemented using a finite state machine (FSM). Any controller can be implemented using an FSM, which is a mathematical model of computation used to design both computer programs and sequential logic circuits. FSM is conceived as an abstract machine that can be used in one of a finite number of states. It can change from one state to another when initiated by a triggering event or condition referred to as transition. FSMs can be classified as Mealy FSM and Moore FSM. In a Mealy FSM, the output depends on the current input and the current state (Fig. 6 (a)) whereas in a Moore FSM, the output depends on the current state only (Fig. 6 (b)).

VHDL code listing

The VHDL code (I2C.vhd) for implementing the I²C master controller is included in this month's EFY DVD. The slave device used here has the 7-bit address of 1001000. The designer should change the address mentioned in the datasheet of the device being used. Also, the code implements the write and the read operations for one byte of data from the slave. The designer can modify the code for writing/reading multiple bytes accordingly. The code is for single master-slave configuration.

VHDL code description

The I2C.vhd code begins with the standard library to be included. The entity section defines the interface between the I²C controller and the outside world. It includes all the input and output connections including the 50MHz clock (CLK_50MHz) and reset (Reset) as input, serial clock (SCL) as output and serial data (SDA) as input-output. SDA is an open-drain bidirectional line. Therefore in the entity declaration, SDA signal is defined as in/out so that it is capable of both input and output operations.

The architecture section defines the operations of the I²C controller and comprises two processes, namely, the *Output* and the *CLK200kHz*. The *Output* process implements the FSM for the I²C controller and the *CLK200kHz* process generates the 200kHz clock from the input clock of 50MHz. The SCL clock is derived from the 200kHz clock.

The different signals defined in the architecture section include *STATE*, *DATA_IN* and *DATA_OUT* which are all 8-bit standard logic vectors, *CLK_200kHz* which is a single-bit standard logic signal and *BITCOUNT*



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